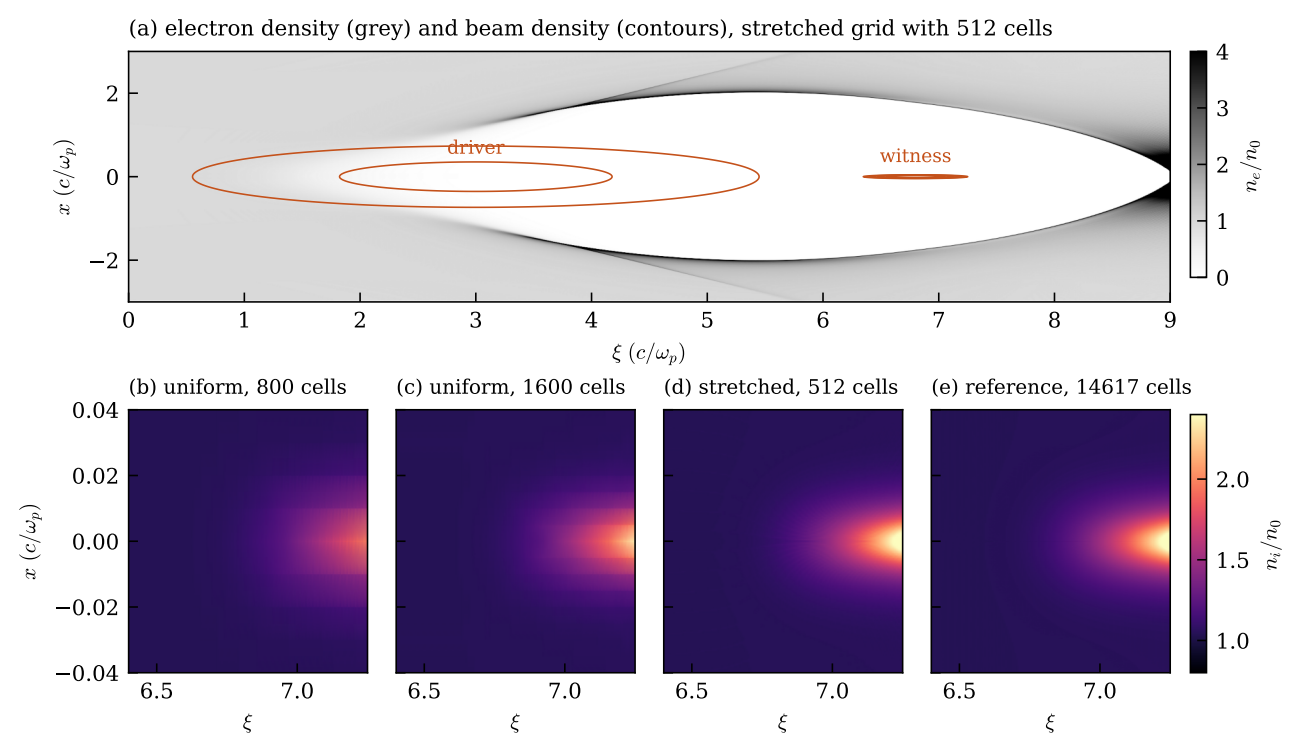

(a) electron density (grey) and beam density (contours), stretched grid with 512 cells
driver
witness
$x\ (c/\omega_p)$
$\xi\ (c/\omega_p)$
n_e/n_0
(b) uniform, 800 cells
(c) uniform, 1600 cells
(d) stretched, 512 cells
(e) reference, 14617 cells
n_i/n_0